

Education

ETH Zürich MSc in Computer Science

Sep. 2024 - Exp. Mar. 2027

- **Major:** Secure & Reliable Systems; **Minor:** Systems Software; GPA **5.68/6.0**;
- **Relevant courses:** Operating Systems, Compiler Design, Computer Architecture, Hardware Security, HPC, Advanced Systems Lab, Digital Circuits

Delft University of Technology BSc in Computer Science

Sep. 2021 - Jun. 2024

- Graduated “with distinction” & Honors (**Competitive Programming track**); GPA: **9.08 / 10.0** (top 2%)
- Exchange semester at **University of Illinois at Urbana-Champaign (UIUC)**; GPA: **4.0 / 4.0** (Dean's List)
- **Relevant courses:** DSA, Operating Systems (Ranked first in class), Systems Programming (A+), Advanced Compute Architecture (A+), Computer Organisation, Computer Networks, Computer Security, Functional Programming, Databases, OOP
- **Awards:** 1 of 10 students to receive **Holland Scholarship**, a merit-based study abroad (exchange) grant.
- **Bachelor's thesis (9/10):** “Investigating the Amplification Potential of Common UDP-Based Protocols in DDoS Attacks across Belgium and Luxembourg”

Professional Experience

OPTIVER - SWE INTERN | C++, VHDL, EMBEDDED SYSTEMS, COMPUTER ARCHITECTURE, OPERATING SYSTEMS

Jun. 2026 - Sep. 2026

- In the Low Latency Engineering team, wrote a **user-space driver** for an SoC's DMA engine in **C++23** and migrated build & flashing off a legacy tool
- **Received return offer** as a new grad for 2027; Ranked **top 10%** of the interns cohort, based on my performance.

SAFARI RESEARCH GROUP (ETH ZURICH) - RESEARCH INTERN | COMPUTER ARCHITECTURE, OPERATING SYSTEMS, C++, RESEARCH

Jun. 2025 - Aug. 2025

- **Fixed** correctness & performance bugs in an imitation-based OS simulator using **C++17 (Virtuoso)**. **Contributed** to a paper rebuttal by implementing additional comparison baselines in the simulator and conducting large-scale experiments to demonstrate the performance advantages of our approach.
- **Refactored** the simulator's infrastructure to enable lock-free communication between the userspace-OS from **Virtuoso** and the simulated application.
- **Leading** an ongoing research project on a new VM scheme that enables the adoption of new optimisations, using a reconfigurable hardware/OS interface; **Implementing and validating** a hardware-accelerated page fault handling mechanism in simulation

BENDING SPOONS - SWE INTERN (BACKEND & PLATFORM) | PYTHON, MYSQL, GCP, TERRAFORM, GRAFANA, DOCKER, CIRCLECI

Jul. 2024 - Sep. 2024

- Performed a 1-month training, receiving positive feedback on performance and self-managing tasks: **good sense of urgency, fast learner**
- Migrated different functionalities from **iTranslate** (Time-To-Speech and Vision feature) into Bending Spoons' infrastructure: rewrote code in **Python**, configured infrastructure using **Terraform** and deployed the backend code in a **Docker container** on **GCP**; created **Grafana dashboards and alerts** that I continuously monitored to improve user experience; ensured seamless integration by constantly communicating with the Frontend team
- Reversed engineered and then migrated Coloring Book app by removing all the backend code, and instead, set up **load balancer** using Fastly CDN (through a Terraform module), **SSL certificates** and **DNS domains** to *completely remove maintenance costs*

GOOGLE ARM SUMMER SCHOOL | C, EMBEDDED SYSTEMS, OPERATING SYSTEMS, BOOTLOADERS, KERNEL DEVELOPMENT, SHELL

Aug. 2024

- **1 of 20 students** selected to participate in a 1-week camp gaining hands-on experience in *embedded systems engineering using ARM microprocessors*
- Focused on the **SoC (System-on-Chip)** architecture from ARM and its integration with the low-level firmware by programming an **NXP IMX8M board**
- Explored the **boot process**, by configuring and compiling its various components from scratch; Got introduced to TEEs and **Linux kernel development**

SOURCER - SWE INTERN | PYTHON, DJANGO, MONGODB, JAVASCRIPT, REACT, BOOTSTRAP CSS, MYSQL, HEROKU, DATA MINING

Apr. 2023 - Jul. 2023

- Developed a **full-stack application** for **Sourcer**, using **Django** on the backend and **React** on frontend, that identifies *text overlapping in the context of news articles*; addressed this issue by combining *state-of-the-art Data Mining algorithms* (Inverted Index and winnowing)
- Managed the database system using **MongoDB** and later migrated to **PostgreSQL** on **AWS** for scalability (capable of handling *100k documents*) and noticeable performance benefits; reduced memory consumption by a significant factor and improved queries' latency by *6x*

Projects

MULTIKERNEL OPERATING SYSTEM | C, OPERATING SYSTEMS, NETWORKING, COMPUTER ARCHITECTURE, SOFTWARE DESIGN

Feb. 2025 - May 2025

- Worked in a team of five towards building a **fully-functional operating system** on top of the **Barrelfish Multikernel**, by understanding the design and implementation issues of modern OSes (emphasising efficient utilisation of modern hardware features)
- Individually implemented an **UART device driver**, a **user-space physical memory allocator**, and the **networking stack** (interacted with the hardware queues to receive and send raw packets, implemented ARP, Ethernet, IP, UDP, and measured performance to shed light on future optimisations).
- Jointly developed core OS functionalities, including **virtual memory management** (paging, page-fault handling), **inter-process communication** (RPC), **process spawning**, and **multi-core bootstrapping**, achieving seamless inter-core coordination and resource sharing

HFT SYSTEM IN C++ | C++, NETWORKING, UDP, BOOST.ASIO, QUANTITATIVE FINANCE

Aug. 2023 - Dec. 2023

- In the Software team of the **Quant Illinois** student association, implemented the **UDP communication** between the feed handler and the ticker plant
- Used **Boost.Asio C++** library to implement UDP; Deserialised the market data on the feed handler according to the Nasdaq's ITCH 5.0 specifications

TCP/IP & UDP PROTOCOLS IN C | C, SYSTEMS PROGRAMMING, NETWORKING, UNIX, MULTITHREADING, IO

Jul. 2023 - Nov. 2023

- Implemented a **CLI chat application** using **C** in the **UNIX environment** by setting up a **multi-threaded TCP/IP client-server**
- Ensured robustness and memory safety through extensive testing, including remote connections via **SSH** and **Valgrind** analysis
- Enhanced the application's scalability and responsiveness by transitioning from a multi-threaded implementation to a **non-blocking I/O model**

HONOURS COMPETITIVE PROGRAMMING RESEARCH PROJECT | C++, ALGORITHM DESIGN, PROBLEM SOLVING, RESEARCH

Jan. 2023 - Jun. 2024

- **Admitted as 1 of 20 students** based on a motivation statement and academic performances; Constructed an **optimal decision tree** in **C++** based on several heuristics to prune the search space; Identified performance bottlenecks and speeded up my solution by *3x* by implementing the **MurTree** paper

RAY TRACING - COMPUTER GRAPHICS | C++, OPENGL, DATA STRUCTURES, ALGORITHM DESIGN

Sep. 2022 - Nov. 2022

- Worked in a team of three to implement a fully functional **recursive ray-tracer** in **C++**; Scored a **perfect 10/10** on the project
- Improved the rendering time by **75%** after implementing **Bounding Volume Hierarchy (BVH)** and enhanced rendering performances by applying Surface Area Heuristic (SAH); Implemented **depth of field (DoF)** for a more realistic effect, and **mipmaps** to improve texture cache efficiency